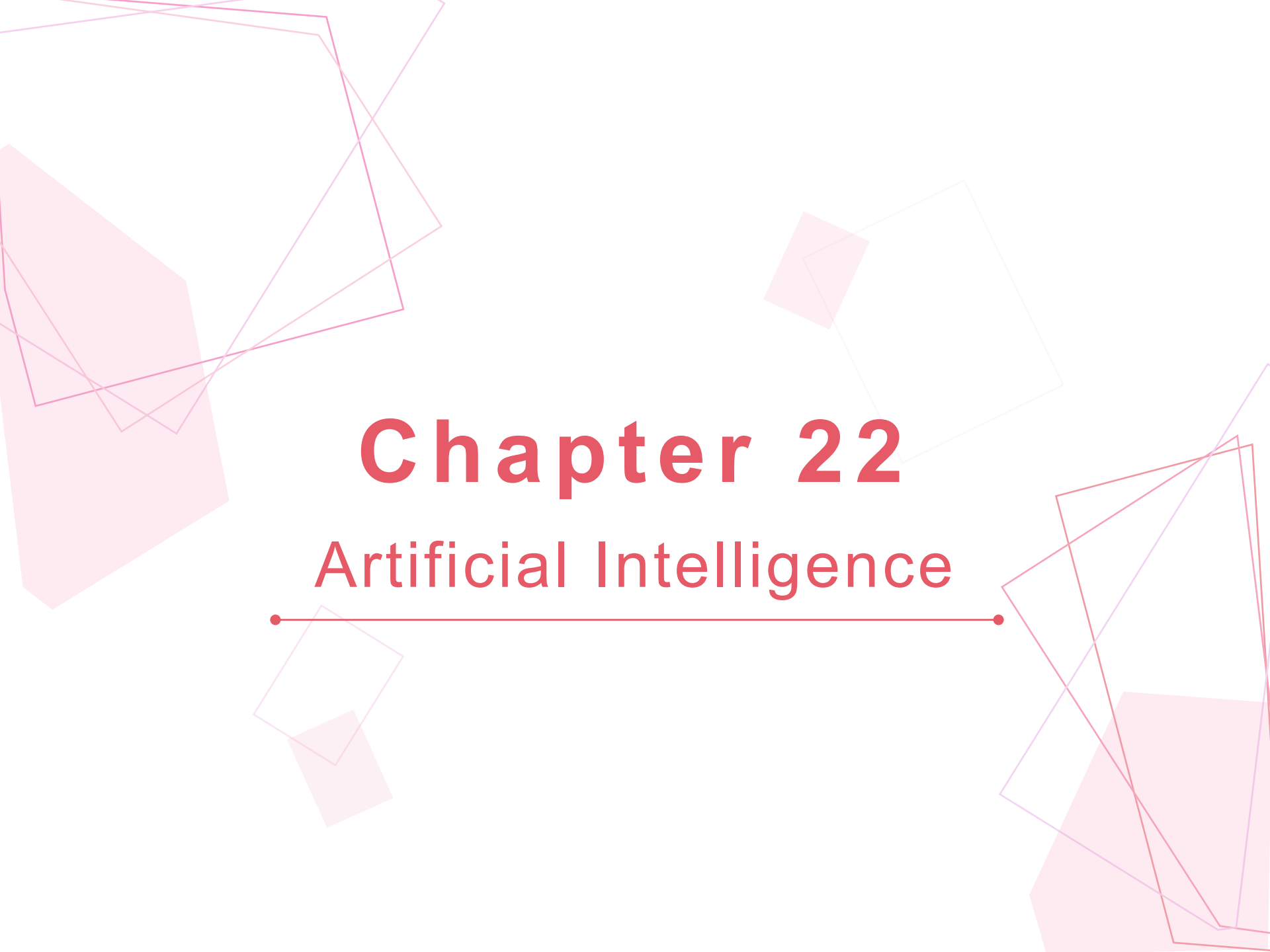




Chapter 22

Artificial Intelligence

Requirements in syllabus

18 Artificial Intelligence (AI)

18.1 Artificial Intelligence (AI)

Candidates should be able to:

Show understanding of how graphs can be used to aid Artificial Intelligence (AI)

Show understanding of how artificial neural networks have helped with machine learning

Show understanding of Deep Learning, Machine Learning and Reinforcement Learning and the reasons for using these methods.

Show understanding of back propagation of errors and regression methods in machine learning

Notes and guidance

Purpose and structure of a graph

Use A* and Dijkstra's algorithms to perform searches on a graph

Candidates will not be required to write algorithms to set up, access, or perform searches on graphs

Understand machine learning categories, including supervised learning, unsupervised learning

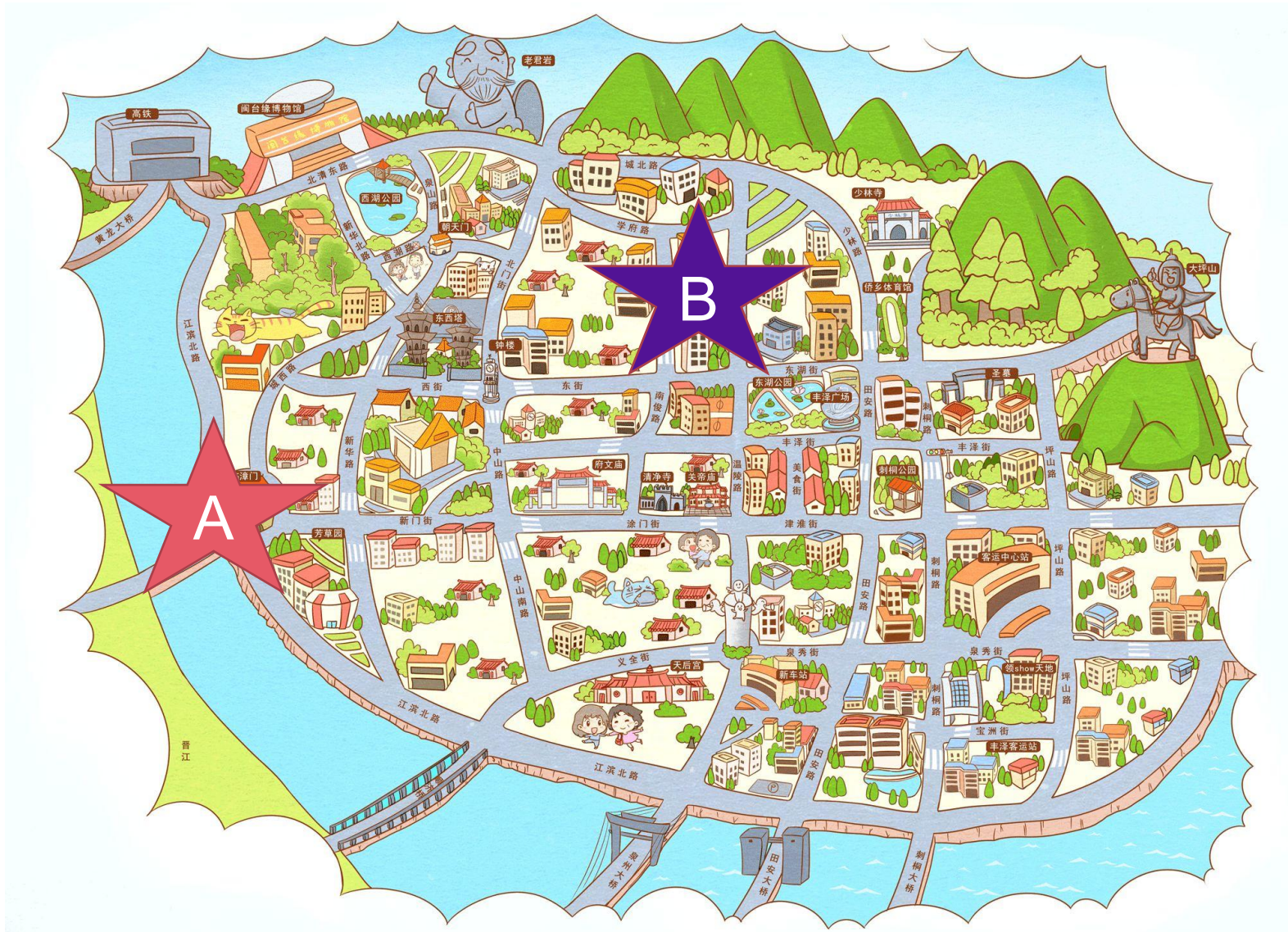
What is AI?

- Artificial Intelligence is concerned with “how to make computers do things at which, at the moment, people are better.”

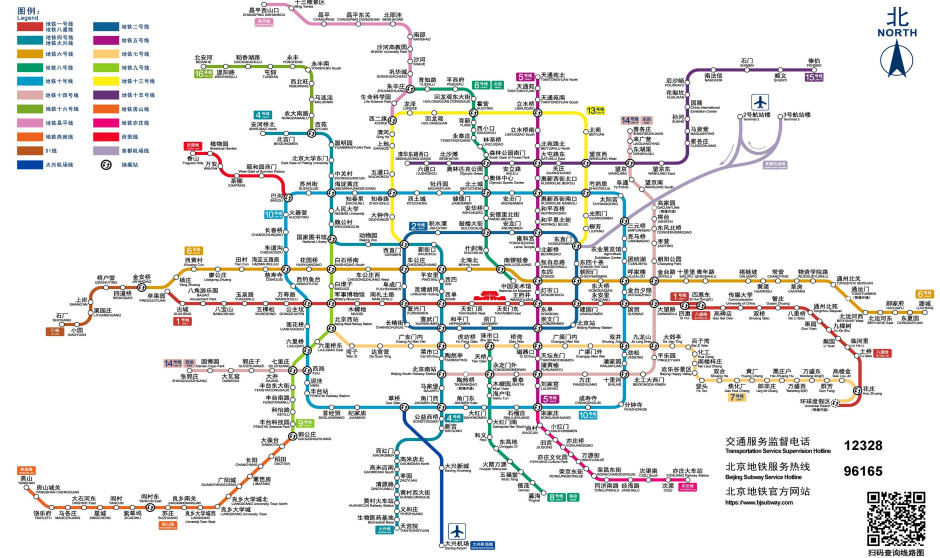
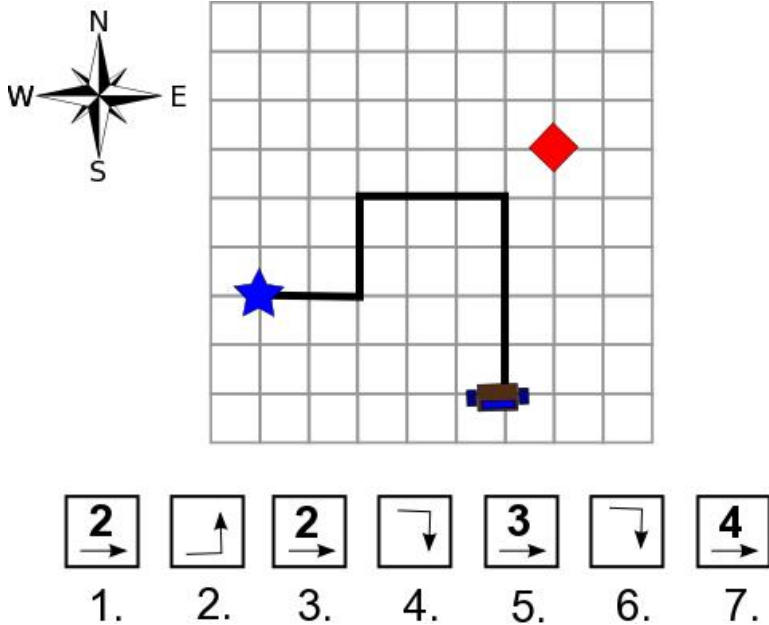
-----E.Rich.Artificial Intelligence. McGraw-Hill, 1983)



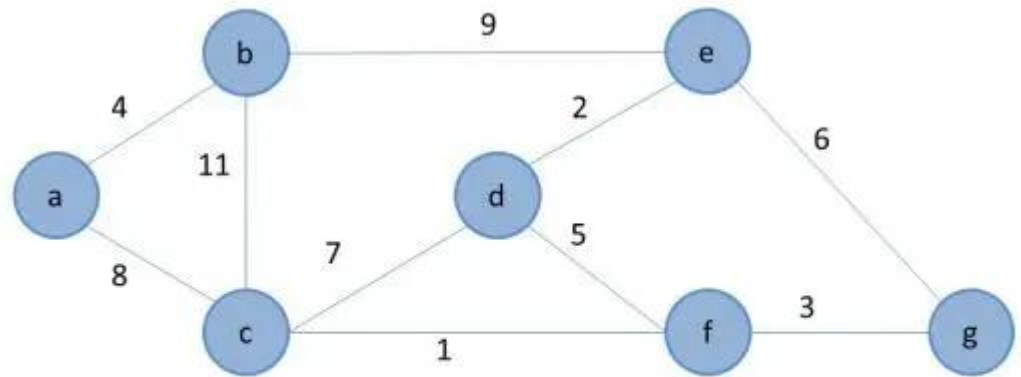
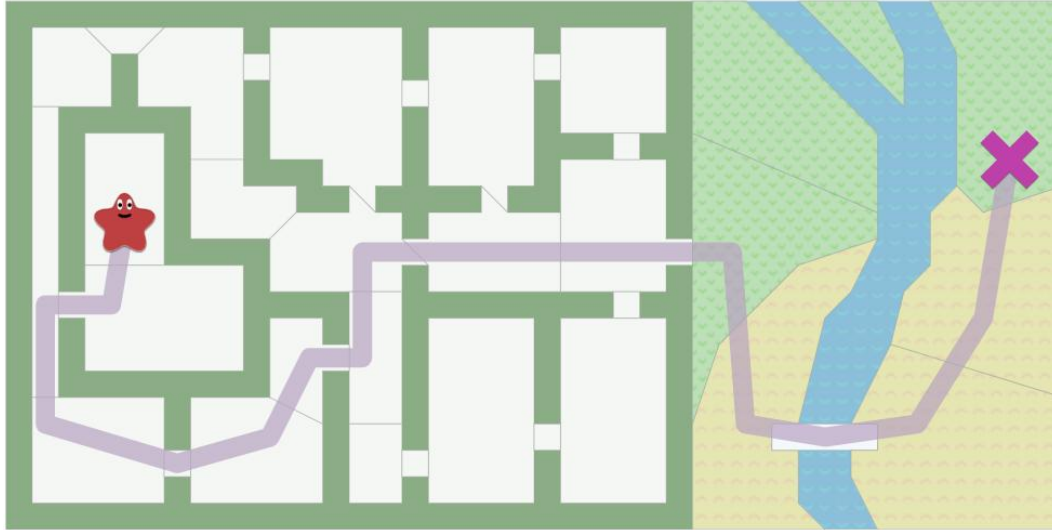
Question: How do a self-driving car find a way from A to B?



Question: How do a self-driving car find a way from A to B?

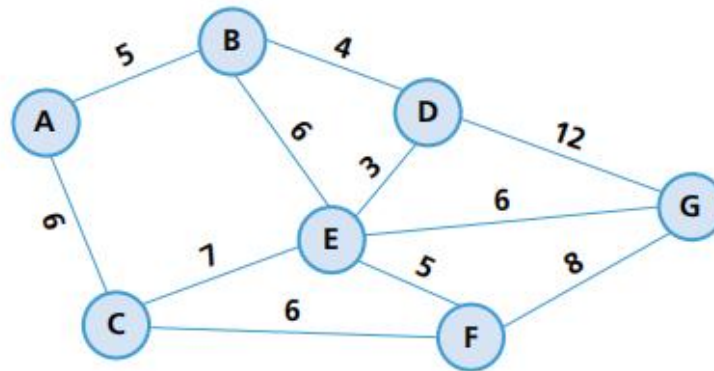


Question: How do a self-driving car find the shortest path ?



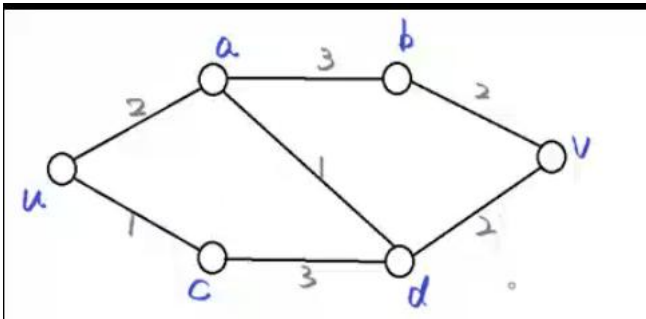
Shortest Path algorithm

Graph: In computer science, a graph is an ADT consisting of **vertices** (nodes) and **edges**. Graphs are used to record relationships between things.



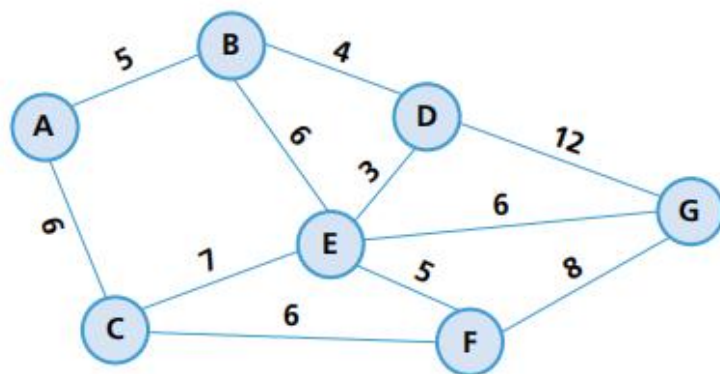
Dijkstra's algorithm

Watch the video and practice



distination	path	distance
a		
b		
c		
d		
v		

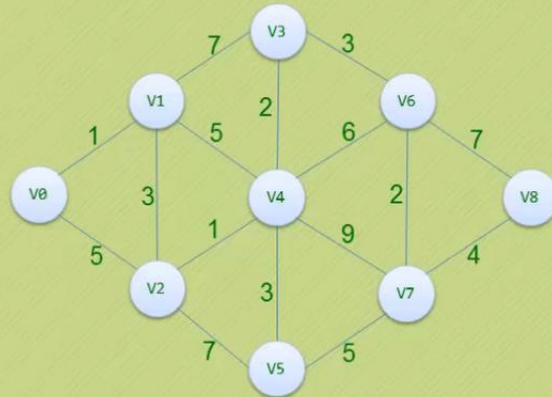
Dijkstra's algorithm



distination	path	distance
B		
C		
D		
E		
F		
G		

Dijkstra's algorithm

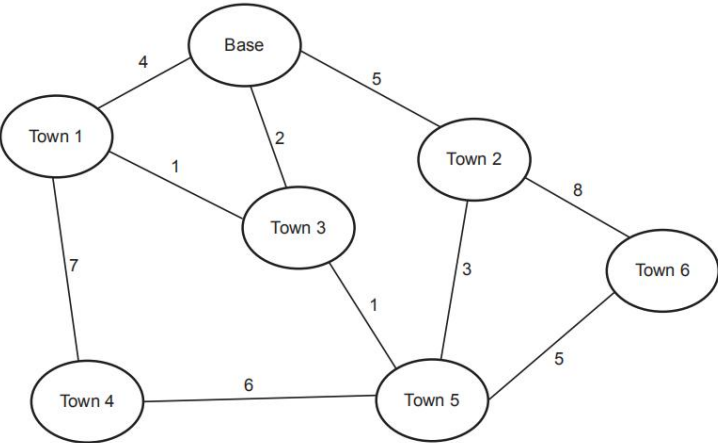
求V0到V8的最短路径



distination	path	distance
V1		
V2		
V3		
V4		
V5		
V6		
V7		
V8		

Past Paper Question: 2021 32 Q5

Calculate the shortest distance between the base and each of the other towns in the diagram using Dijkstra's algorithm.
Show your working and write your answers in the table provided.



Working

.....

.....

.....

.....

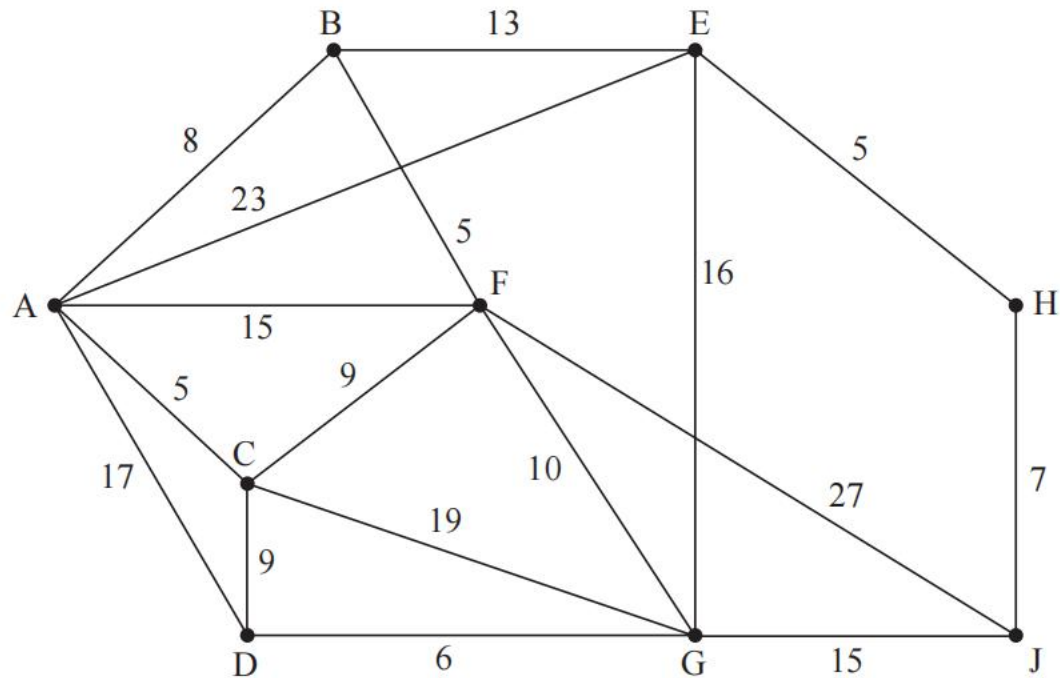
.....

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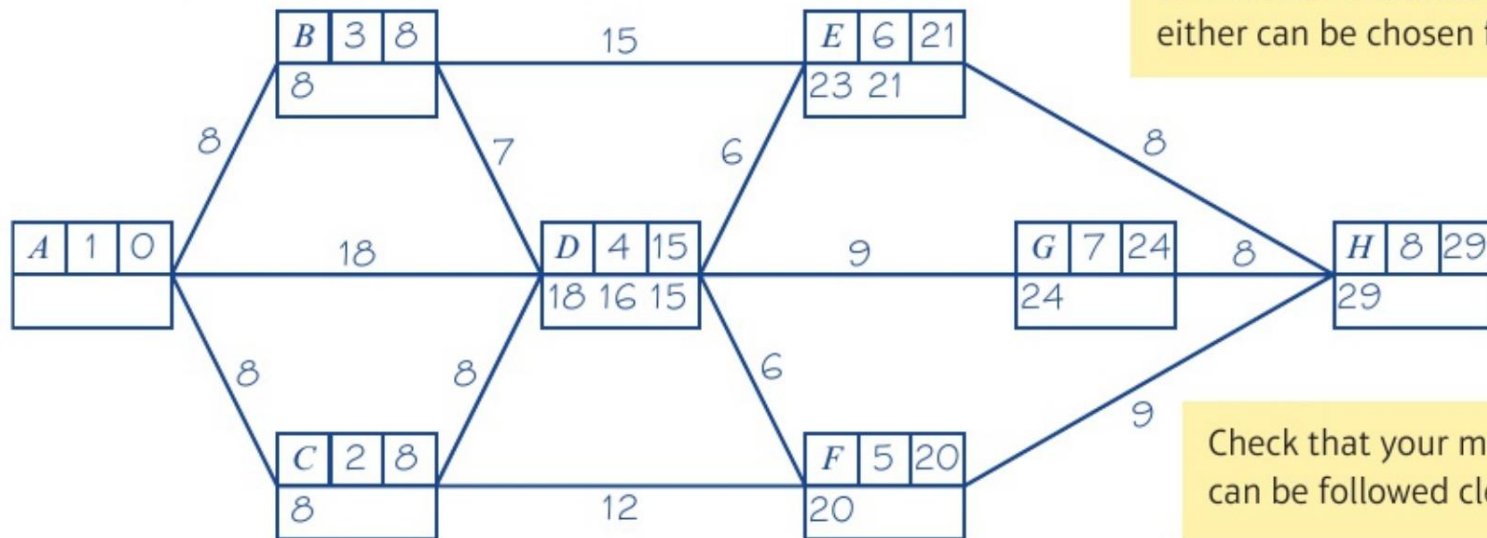
Answers

Town 1	Town 2	Town 3	Town 4	Town 5	Town 6

Using Dijkstra's algorithm



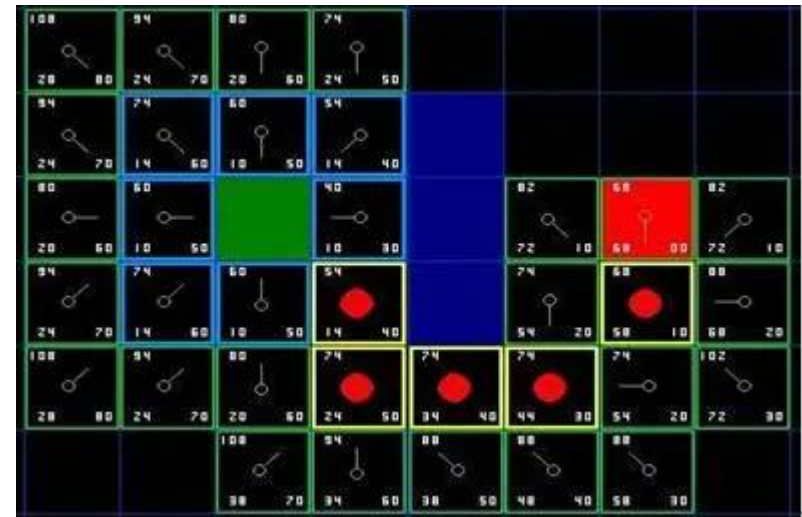
- a The final diagram looks like this. It shows all the working needed to make the method clear.



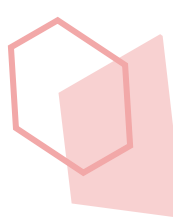
The lowest working values at *B* and at *C* are both 8, so either can be chosen first.

Check that your method can be followed clearly.

efficiently.

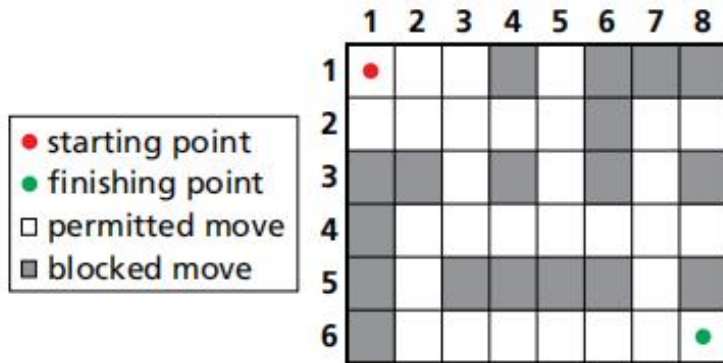


Watch the video



A* algorithm

Suppose we have the following graph made up of an 8×6 matrix. White squares show permitted moves, and grey squares show blocked moves.

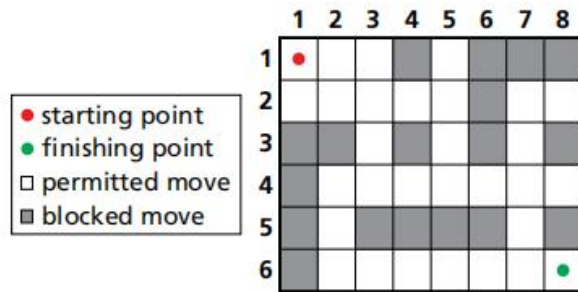


Each of the parts of the graph are called nodes (or vertices). Each node has four values

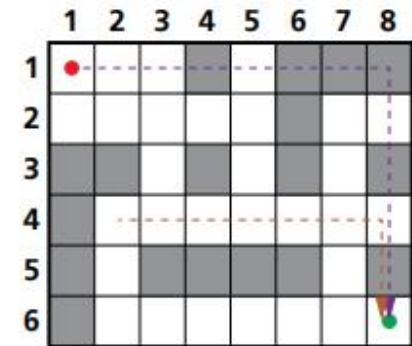
- » h (the heuristic value)
- » g (movement cost)
- » f (sum of g and h values)
- » n (previous node in the path).

A* algorithm

Suppose we have the following graph made up of an 8×6 matrix. White squares show permitted moves, and grey squares show blocked moves.



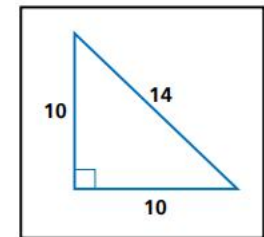
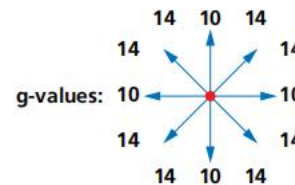
Heuristic value: Manhattan Distance



Each of the parts of the graph are called nodes (or vertices). Each node has four values

- » h (the heuristic value)
- » g (movement cost)
- » f (sum of g and h values)
- » n (previous node in the path).

g-values (the movement costs).



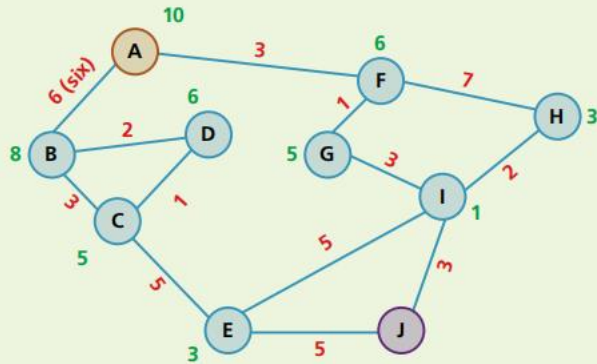
ACTIVITY 18B

1 Find the shortest route from node A to node J using the A* algorithm. Note that you are given the heuristic values (h) in green, and the movement cost values (g), in red.



```
graph LR; A((A)) ---|3| B(( )); B ---|6| C(( ))
```

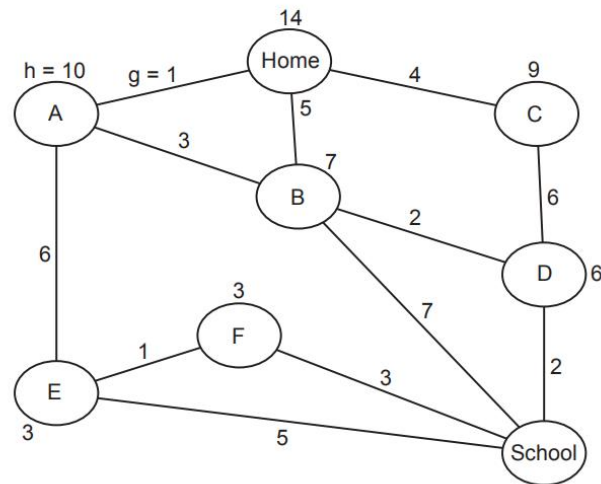
- Find the shortest route from node A to node J using the A* algorithm. Note that you are given the heuristic values (h) in green, and the movement cost values (g), in red.

[illegible]

Past Paper Question: 2021 W 31/32 Q9 b

(b) Find the shortest path between the Home and School nodes using the A* algorithm.
Show your working in the table provided.

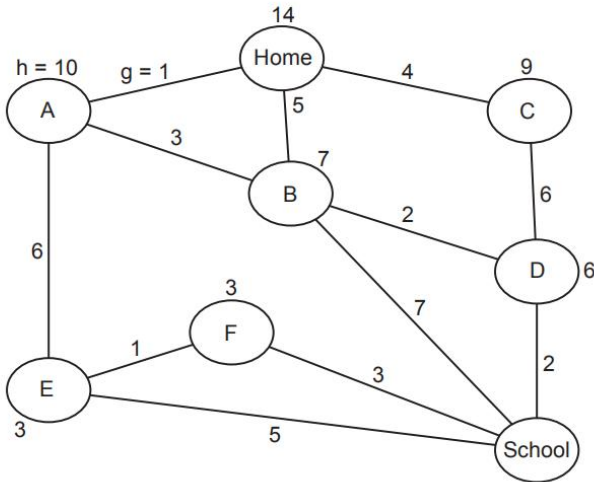
The first two rows in the table have been completed.



Node	Cost from Home node (g)	Heuristic (h)	Total (f = g + h)
Home	0	14	14
A	1	10	11

- (b) Find the shortest path between the Home and School nodes using the A* algorithm. Show your working in the table provided.

The first two rows in the table have been completed.



Node	Cost from Home node (g)	Heuristic (h)	Total (f = g + h)
Home	0	14	14
A	1	10	11

One mark for each correct calculation as follows (Max 4)

- Node B (from Home) (Line 3 in table)
- Node C (from Home) (Line 4 in table)
- Node B and Node E (from A) (Lines 5 and 6 in table)
- Node F and Node School (from E) (Lines 7 and 8 in table)
- Node School (from F) (Line 9 in table)

One mark for correct path (Max 1):

- Home $\Rightarrow$ A $\Rightarrow$ E $\Rightarrow$ F $\Rightarrow$ School

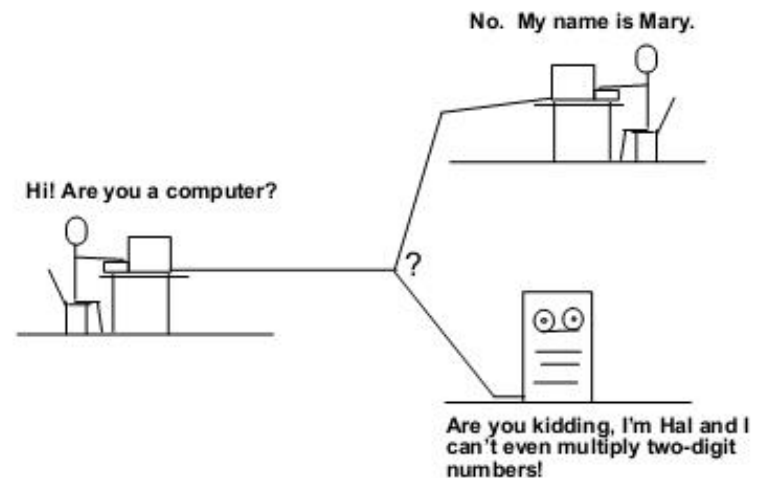
	Node	Cost from Home Node (g)	Heuristic (h)	Total (f = g + h)
1	Home	0	14	14
2	A	1	10	11
3	B	5	7	12
4	C	4	9	13
5	B	1 + 3 = 4	7	11
6	E	1 + 6 = 7	3	10
7	F	7 + 1 = 8	3	11
8	School	7 + 5 = 12	0	12
9	School	8 + 3 = 11	0	11

Final Path	Home $\Rightarrow$ A $\Rightarrow$ E $\Rightarrow$ F $\Rightarrow$ School
------------	---

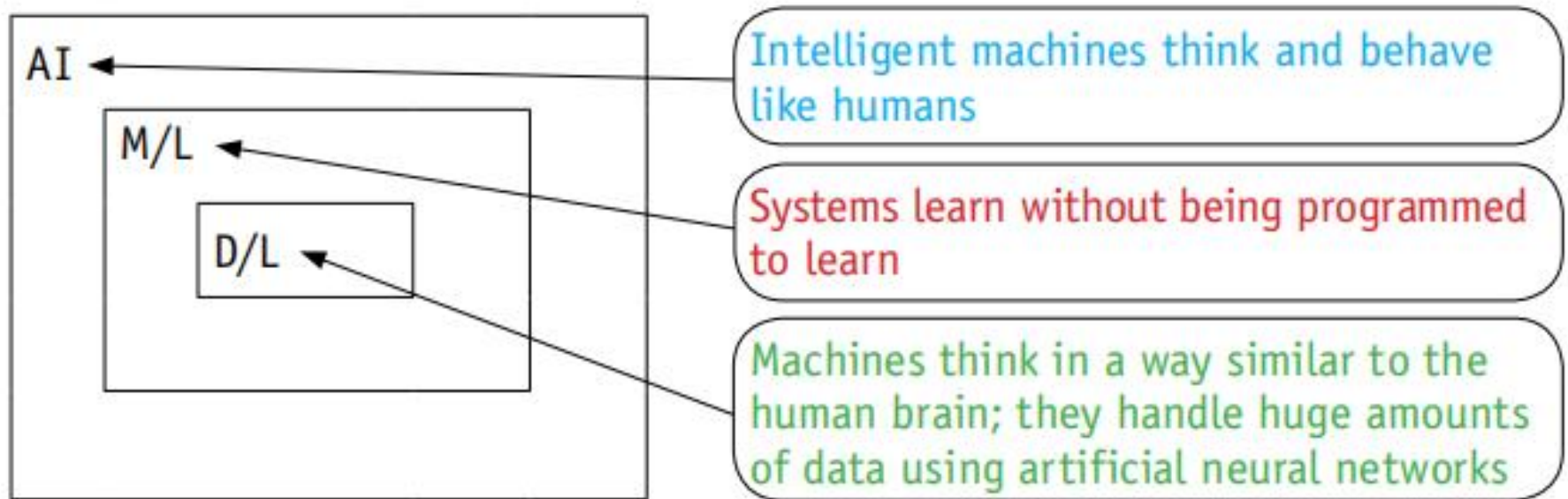
Types of AI

1. **Narrow AI** is when a machine has superior performance to a human when doing one specific task.
2. **General AI** is when a machine is similar in its performance to a human in any intellectual task.
3. **Strong AI** is when a machine has superior performance to a human in many tasks.

A Turing Test is a method of inquiry in artificial intelligence (AI) for determining whether or not a computer is capable of thinking like a human being. The test is named after Alan Turing, the founder of the Turing Test and an English computer scientist, cryptanalyst, mathematician and theoretical biologist.



AI and Machine Learning



Machine learning

How do we say a machine can learn?

Machine learning is a subset of AI, in which the algorithms are ‘trained’ and learn from their past experiences and examples. It is possible for the system to make predictions or even take decisions based on previous scenarios.

One of the definition:

A computer program is said to learn from experience E with respect to some task T and some performance measure P . If its performance on T , as measured by P , improves with experience E .

Machine Learning

Example 1:

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. Which of the following is T,E,P.

- A. Classify emails as spam or not spam
- B. Watching you label emails as spam or not spam
- C. The number(or fraction) of emails correctly classified as spam.

How do we say a machine can learn?

One of the definition:

A computer program is said to learn from experience E with respect to some

task T and some performance measure P. If its performance on T, as measured by P, improves with experience E.

Machine Learning

Example 2:

Suppose your program is to predict the housing price in a area based on the exit data. Which of the following is T,E,P.

- A. Predict the housing price
- B. find the function to calculate the price base on the given data
- C. how much is the prediction closed to the real price

Types of machine learning

There are a number of different types of machine learning, including :

supervised, **unsupervised learning**, **reinforcement** and **semi-supervised**

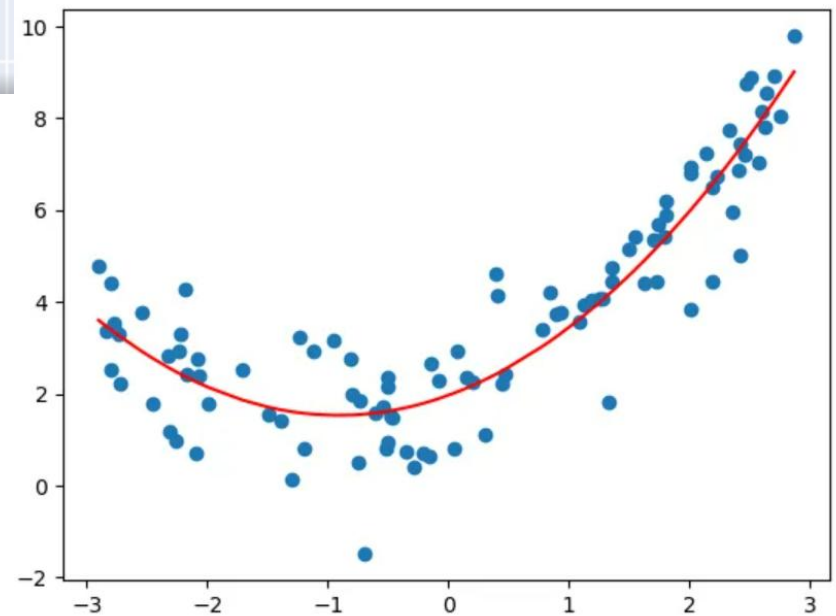
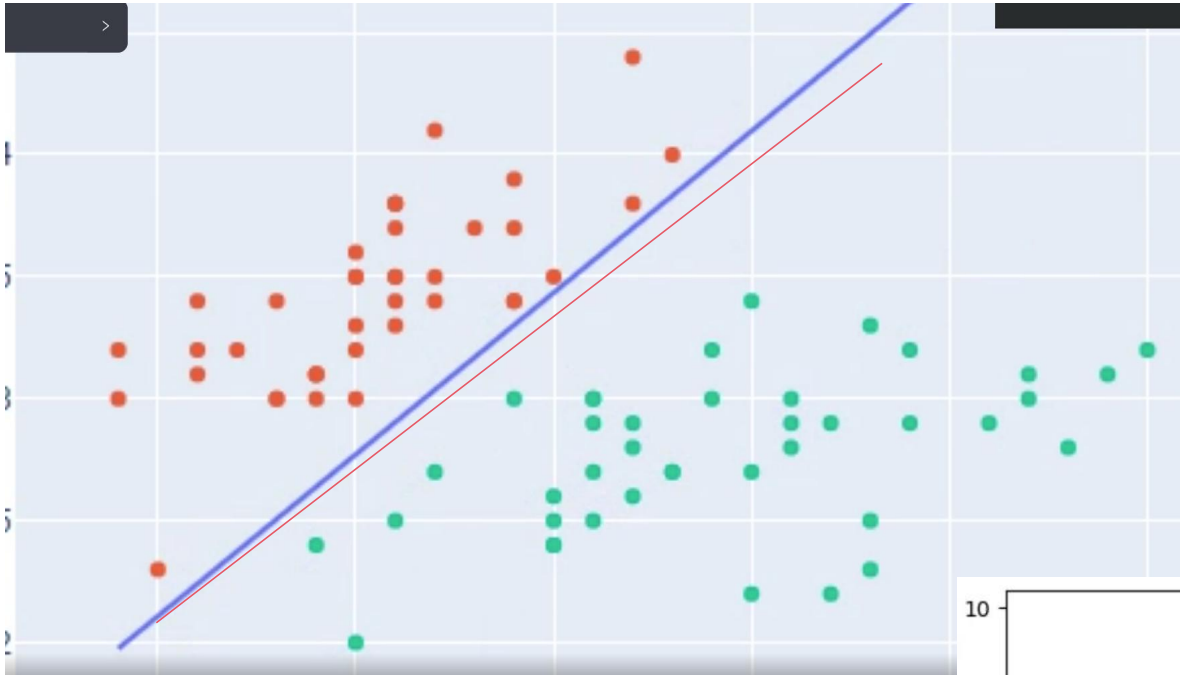
Supervised learning

Supervised learning makes use of regression analysis and classification analysis. It is used to predict future outcomes based on past data:

- » The system requires both an input and an output to be given to the model so it can be trained.
- » The model uses labelled data, so the desired output for a given input is known.
- » Algorithms receive a set of inputs and the correct outputs to permit the learning process.
- » Once trained, the model is run using **labelled data**.
- » The results are compared with the expected output; if there are any errors, the model needs further refinement.
- » The model is run with unlabelled data to predict the outcome.

An example of supervised learning is categorising emails as relevant or spam/junk without human intervention.

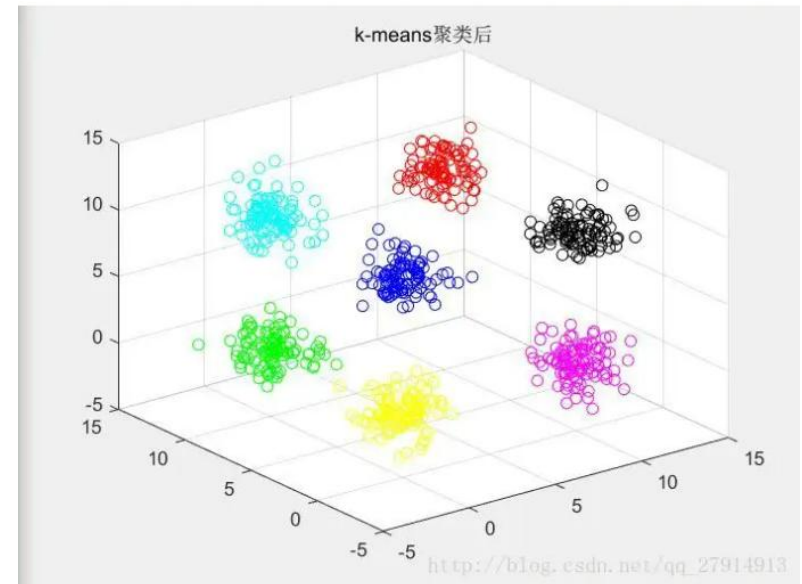
Classification and Regression



Unsupervised learning

Systems are able to identify hidden patterns from the input data provided; they are not trained using the ‘right’ answer.

By making data more readable and more organised, patterns, similarities and anomalies will become evident (unsupervised learning makes use of density estimation(密度估计) and k-mean clustering (k聚类); in other words, it classifies unlabelled real data).



“你，就是你最常接触的五个人的平均。”

An example is used in product marketing: a group of individuals with similar purchasing behaviour are regarded as a single unit for promotions.

Reinforcement learning

The system is not trained. It learns on the basis of ‘reward and punishment’ when carrying out an action (in other words, it uses trial and error in algorithms to determine which action gives the highest/optimal outcome).

This type of learning helps to increase the efficiency of the system by making use of optimisation techniques.

Examples include search engines, online games and robotics.

Semi-supervised (active) learning

Semi-supervised learning makes use of labelled and unlabelled data to train algorithms that can interactively query the source data and produce a desired output.

It makes as much use of unlabelled data as possible (this is for cost reasons, since unlabelled data is less expensive than labelled data when carrying out data categorisation).

A small amount of labelled data is used combined with large amounts of unlabelled data.

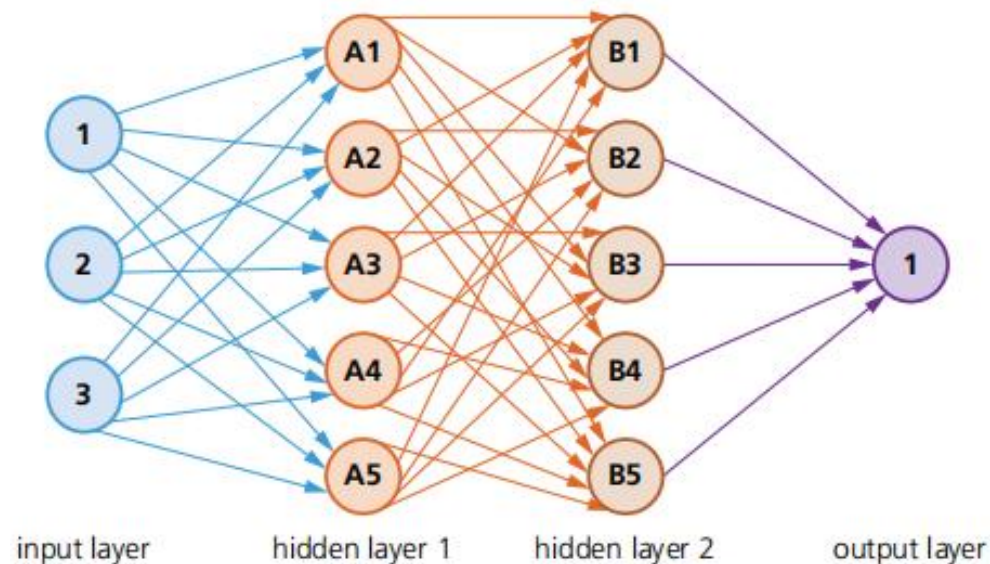
Examples of uses include the classification of web pages into sport, science, leisure, finance, and so on. A web crawler is used to look at large amounts of unlabelled web pages, which is much cheaper than going through millions of web pages and manually annotating (labelling) them.

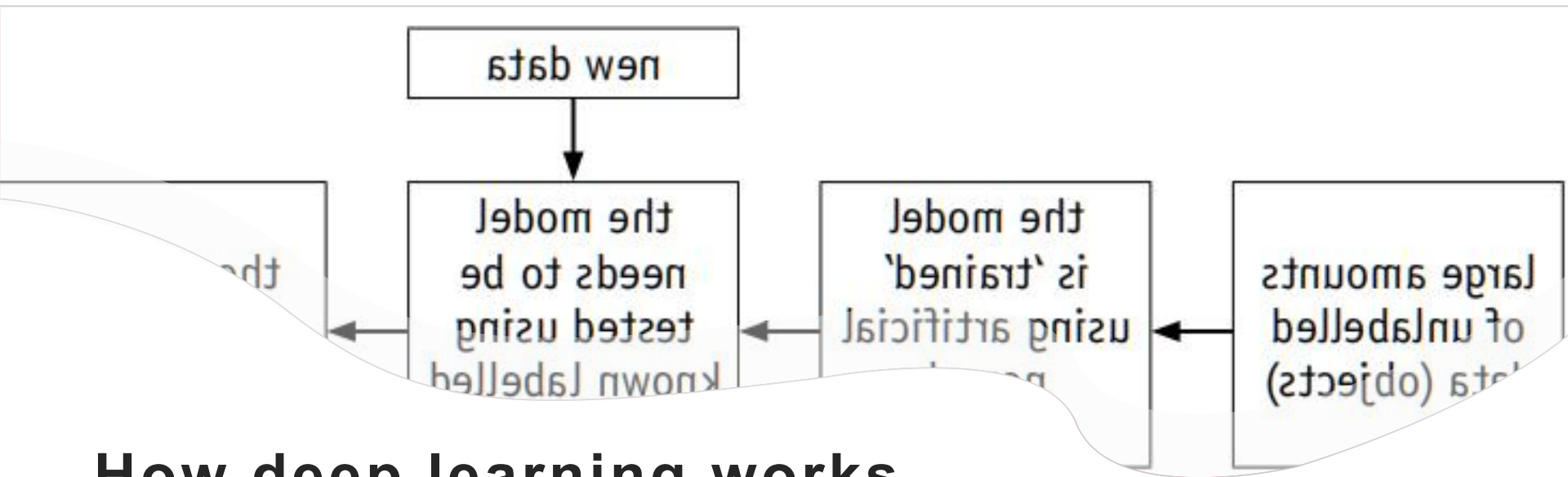
Deep learning

Deep learning structures algorithms in layers (input layer, output layer and hidden layer(s)) to create an artificial neural network that can learn and make intelligent decisions on its own.

Its **artificial neural networks** are based on the interconnections between neurons in the human brain. The system is able to think like a human using these neural networks, and its performance improves with more data.

The hidden layers are where data from the input layer is processed into something which can be sent to the output layer. Artificial neural networks are excellent at identifying patterns which would be too complex or time consuming for humans to carry out.





How deep learning works

<http://playground.tensorflow.org>

<http://neuralnetworksanddeeplearning.com/chap1.html>

Explain deep learning

Deep learning is a subset of machine learning, which in turn is a branch of artificial intelligence (AI) that involves algorithms and models inspired by the structure and function of the human brain's neural networks. The "deep" in deep learning refers to the use of **multiple layers** in these networks—deep learning models consist of an **input layer**, **multiple hidden layers**, and an **output layer**. Each layer contains units, or neurons, that perform operations on the data passing through them. These operations are typically **non-linear transformations**, which enable the model to **learn complex patterns** and representations from data.

Deep learning models are trained by feeding them large amounts of data and using a process called **backpropagation**, where the model's predictions are compared to the actual outcomes, and the difference (or error) is used to adjust the model's parameters. This training process requires significant computational power, often necessitating the use of specialized hardware like GPUs (Graphics Processing Units).

Explain deep learning

Deep learning has been successful in a wide range of applications, including but not limited to:

- **Image recognition and processing**: where models can identify and classify objects within images, or even generate images.
- **Natural language processing (NLP)**: enabling models like ChatGPT to understand, generate, and translate human languages.
- **Speech recognition**: where models transcribe spoken language into text.
- **Autonomous vehicles**: enabling cars to recognize objects, make decisions, and navigate roads independently.
- **Game playing**: allowing models to master complex games like Go or chess.

The key advantage of deep learning over traditional machine learning is its ability to automatically learn and improve from experience without being explicitly programmed with specific rules. This is particularly useful for tasks where the rules are too complex to be manually coded or where the amount of data is too large for traditional algorithms to handle effectively.

Reasons use deep learning

Deep learning has gained immense popularity and utility across various fields due to several compelling reasons:

1. **Ability to Handle Vast Amounts of Data**: Deep learning algorithms excel as the volume of data increases, making them well-suited for the big data era where traditional algorithms might struggle.
2. **Automatic Feature Extraction**: Unlike traditional machine learning models that often require manual feature selection and extraction, deep learning models can automatically learn complex features from raw data, simplifying the preprocessing step and potentially uncovering novel insights.
3. **Improved Accuracy**: As more data becomes available for training, deep learning models tend to improve their performance, often outperforming traditional algorithms in tasks like image and speech recognition, and natural language understanding.
4. **Versatility and Adaptability**: Deep learning models can be applied to a wide range of applications, from computer vision and natural language processing to autonomous vehicles and personalized recommendations, demonstrating their flexibility.
5. **Continuous Learning and Improvement**: With the ability to learn from incremental data, deep learning models can continuously improve their performance over time, making them ideal for applications where new data is constantly generated.

Reasons use deep learning

Deep learning has gained immense popularity and utility across various fields due to several compelling reasons:

6. **Handling Unstructured Data**: Deep learning models are particularly adept at working with unstructured data such as text, images, and audio, which constitute a significant portion of the data generated today.

7. **End-to-End Learning**: Deep learning allows for end-to-end learning where raw data can be input directly into the model, and the model can learn to perform tasks without needing to manually specify intermediate representations or transformations.

8. **Parallel Processing Capability**: Deep learning models, particularly those using convolutional neural networks (CNNs) and recurrent neural networks (RNNs), can be parallelized efficiently on GPUs and TPUs, significantly reducing training and inference times.

9. **Innovations and Community Support**: The field of deep learning is rapidly evolving with continuous research and development, leading to new models, techniques, and applications. This progress is supported by a vibrant community and open-source projects, making advanced tools and software widely available.

10. **Cross-Disciplinary Applications**: Beyond traditional domains, deep learning has been applied successfully in areas like healthcare for drug discovery and disease diagnosis, in finance for fraud detection, and in environmental sciences for climate modeling, showcasing its wide-ranging impact.

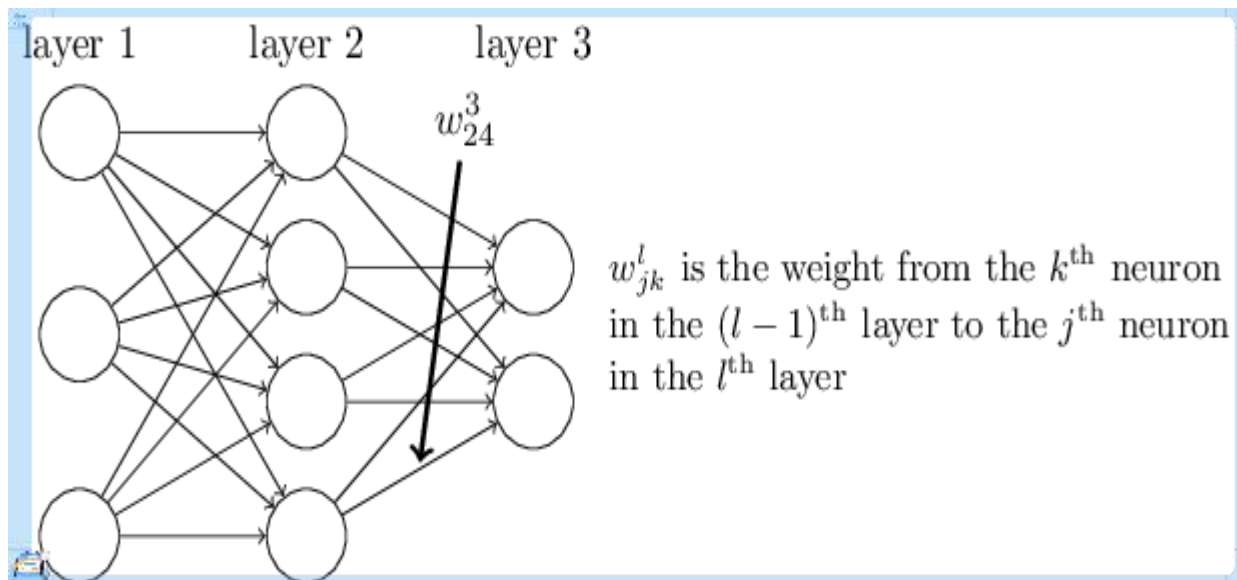
These reasons collectively make deep learning a powerful and attractive choice for tackling complex problems that require insights from large volumes of data across diverse domains.

Machine Learning VS Deep Learning

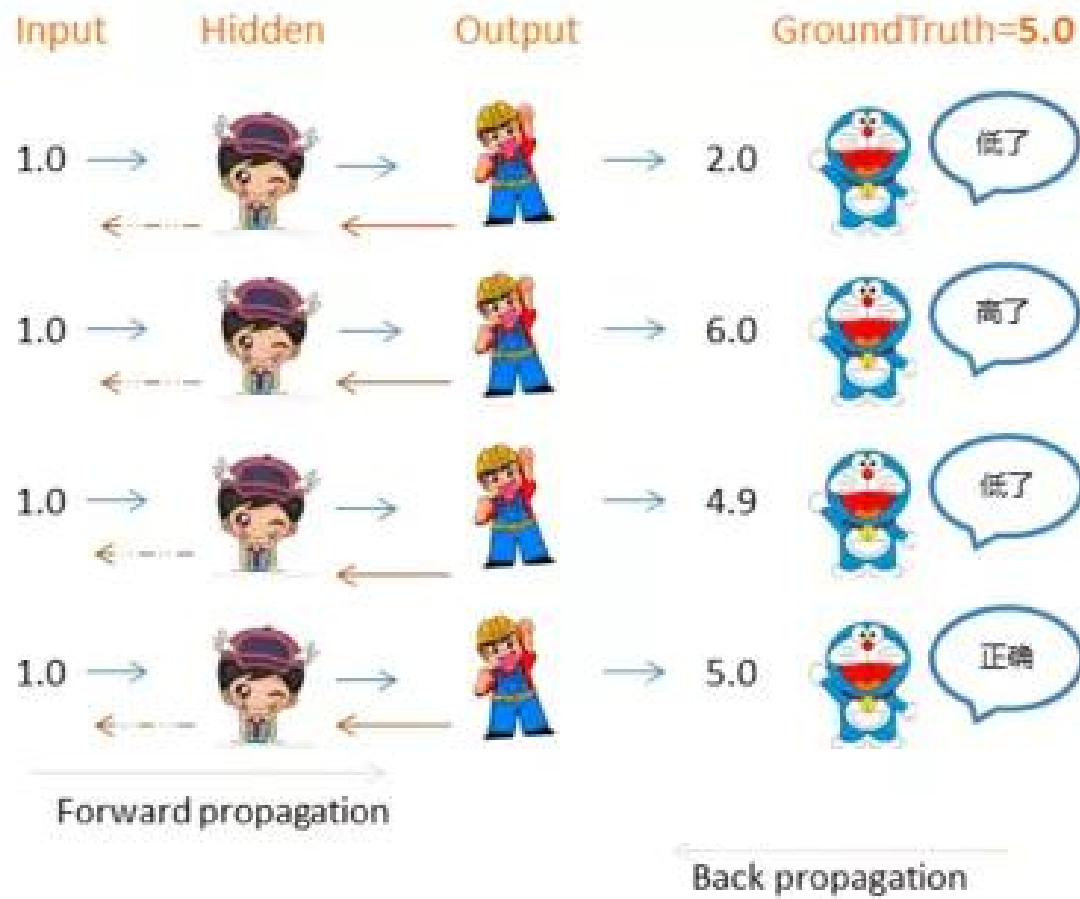
machine learning	deep learning
enables machines to make decisions on their own based on past data	enables machines to make decisions using an artificial neural network
needs only a small amount of data to carry out the training	the system needs large amounts of data during the training stages
most of the features in the data used need to be identified in advance and then manually coded into the system	deep learning machine learns the features of the data from the data itself and it does not need to be identified in advance
a modular approach is taken to solve a given problem/task; each module is then combined to produce the final model	the problem is solved from beginning to end as a single entity
testing of the system takes a long time to carry out	testing of the system takes much less time to carry out
there are clear rules which explain why each stage in the model was made	since the system makes decisions based on its own logic, the reasoning behind those decisions may be very difficult to understand (they are often referred to as a black box)

Back propagation/反向传播

When designing neural networks, it is necessary to give random weightings to each of the neural connections. However, the system designer will not initially know which weight factors to use to produce the best results. It is necessary to train the neural networks during the development stage:



Back propagation



Back propagation

Here is a summary of the back propagation process:

- » The initial outputs from the system are compared to the expected outputs and the system weightings are adjusted to minimise the difference between actual and expected results.
- » Calculus is used to find the error gradient in the obtained outputs: the results are fed back into the neural networks and the weightings on each neuron are adjusted (note: this can be used in both supervised and unsupervised networks).
- » Once the errors in the output have been eliminated (or reduced to acceptable limits) the neural network is functioning correctly and the model has been successfully set up.
- » If the errors are still too large, the weightings are altered – the process continues until satisfactory outputs are produced.

Linear Regression线性回归

square_feet(面积)	price (价格)
150	6450
200	7450
250	8450
300	9450
350	11450
400	15450
600	18450

Question:

How do we predict the price of a house which square feet is 268 in the same area?

Linear Regression线性回归

TV	Radio	Newspaper	Sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	12
151.5	41.3	58.5	16.5
180.8	10.8	58.4	17.9
8.7	48.9	75	7.2
57.5	32.8	23.5	11.8
120.2	19.6	11.6	13.2
8.6	2.1	1	4.8
199.8	2.6	21.2	15.6
66.1	5.8	24.2	12.6
214.7	24	4	17.4
23.8	35.1	65.9	9.2
97.5	7.6	7.2	13.7
204.1	32.9	46	19
195.4	47.7	52.9	22.4
67.8	36.6	114	12.5

Qestion: based on these data, how do you make decision?

how do you find the relationship among advertising costs and sales?

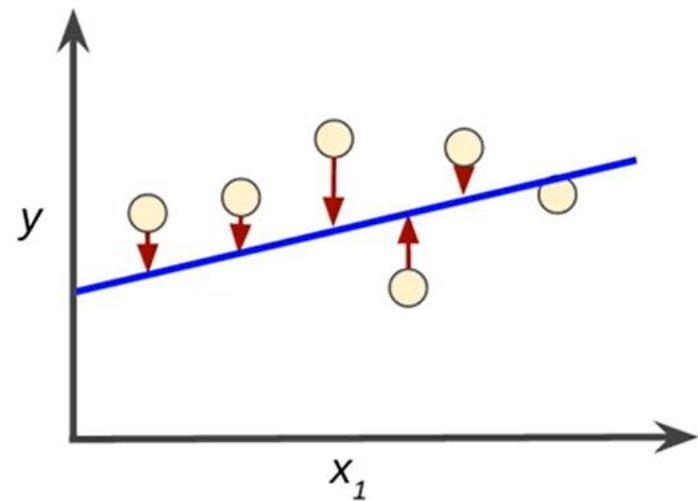
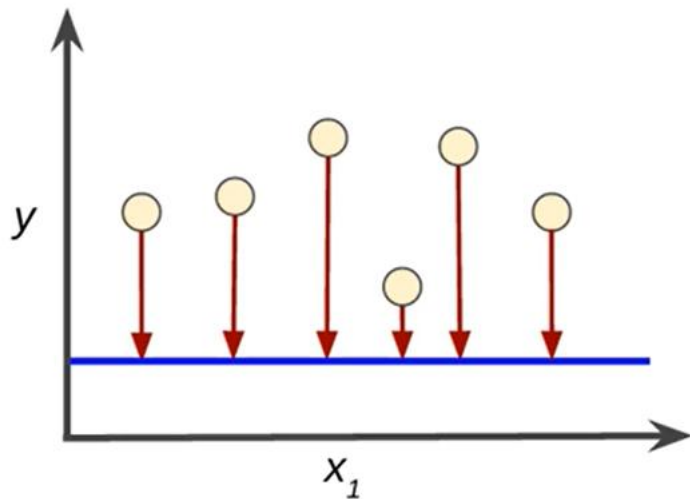
Linear Regression 线性回归

The process of finding a straight line (as by least squares/最小二乘法) that best approximates a set of points on a graph.

“线性回归”试图学得一个线性模型以尽可能准确地预测实值输出标记。即通过学习获得一条线性回归模型，当新的对象进入模型时能够准确的对它进行预测，使得预测值接近真实值。

What is loss?

In mathematical optimization and decision theory, a loss function or cost function is a function that maps an event or values of one or more variables onto a real number intuitively(直观的) representing some "cost" associated with the event. ... In optimal control, the loss is the penalty for failing to achieve a desired value.



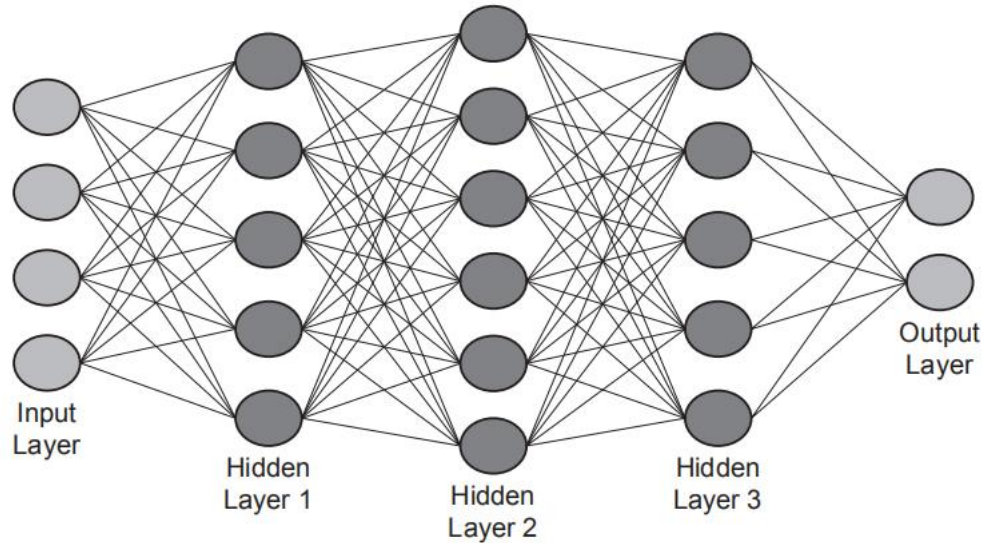
Regression

Regression is used to make predictions from given data by learning some relationship between the input and the output. It helps in the understanding of how the value of a dependent variable changes when the values of independent variables are also changed. This makes it a valuable tool in prediction applications, such as weather forecasting.

In machine learning, this is used to predict the outcome of an event based on any relationship between variables obtained from input data and the hidden parameters.

Past Paper Question: 2021 W 31/32

- 9 (a) The diagram shown represents an artificial neural network.



- (i) State the reason for having multiple hidden layers in an artificial neural network.

.....
..... [1]

- (ii) Explain how artificial neural networks enable machine learning.

.....
.....
..... [4]

9(a)(i)	One mark for correct statement (Max 1) • Enables deep learning to take place • Where the problem you are trying to solve has a higher level of complexity it requires more layers to solve • To enable the neural network to learn and make decisions on its own • To improve the accuracy of the result.	1
9(a)(ii)	One mark for each correct marking point (Max 4) • Artificial neural networks are intended to replicate the way human brains work • Weights / values are assigned for each connection between nodes • The data are input at the input layer and are passed into the system • They are analysed at each subsequent (hidden) layer where characteristics are extracted / outputs are calculated • ... this process of training / learning is repeated many times to achieve optimum outputs // reinforcement learning takes place • Decisions can be made without being specifically programmed • The deep learning net will have created complex feature detectors • The output layer provides the results • Back propagation (of errors) will be used to correct any errors that have been made.	4